

Gavin Allison

Saskatoon, Saskatchewan

CONTACT

✉ gavinallison009@gmail.com

☎ (306) 250-6657

🌐 github.com/Gavin-Allison/

🌐 linkedin.com/in/gavin-allison-/

SKILLS

Programming:

Python, TypeScript, NodeJS, SQL, Java

Web Development:

React, HTML, CSS, Tailwind CSS

Machine Learning:

Pytorch, OpenCV, Numpy, Pandas

EDUCATION

Master of Visual Computing

Simon Fraser University

2025 - Present

B.Sc. Honours Computer Science

University of Saskatchewan

2021 - 2024

WORK EXPERIENCE

Shoppers Drug Mart

Photo Technician

2019 - 2021

- Maintained high-quality service
- Managed digital imaging workflows
- Controlled store inventory levels
- Collaborated efficiently within teams

SUMMARY

Computer Science Masters student specializing in full-stack development, computer vision, and machine learning. Experienced in building ML pipelines, researching object detection and developing applications.

View my work at <https://gavin-allison.github.io/profile>.

PROJECTS

Web App for Investing Backtesting

TypeScript, React, Tailwind CSS, PostgreSQL

- Engineered a TypeScript backtesting algorithm to generate multiple hypothetical stock portfolios for each user
- Developed a responsive front-end with `react-window` for rendering 1000+ transactions with minimal latency
- Designed a PostgreSQL database schema to securely manage distinct user profiles, portfolios, and transaction histories
- Integrated Google GenAI NLP parsing to allow for intuitive transaction entry and greater flexibility over data management.

Object Detection Machine Learning Research Project

Python, PyTorch, Matplotlib, Numpy

- Collaborated with a computer vision expert to quantify the impact of annotation quality and inter-annotator agreement on object detection performance
- Designed an algorithm to automatically perturb bounding boxes based on a normal distribution to simulate annotator skill levels
- Proved a linear deterioration in model performance, measured by a 5% drop in mAP for each level of bounding box perturbation
- Authored and presented a comprehensive research paper on empirical data quality findings to faculty members and peers

Image Classification Segmentation Pipeline

Python, PyTorch, OpenCV, Numpy

- Designed and trained U-Net and ResNet-18 convolutional neural networks in PyTorch for image segmentation and classification on 58000 images
- Built a custom PyTorch Dataset class and automated preprocessing pipeline for image augmentation and mask generation.
- Implemented a categorical parallel training approach, reducing model training time in half while maintaining 90% overall pixel accuracy
- Visualized training metrics and model outputs into interpretable charts to effectively communicate performance improvements